

ANSHU SINGH

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EDUCATION

Vellore Institute of Technology
Bachelor of Technology in Computer Science

Sep. 2023 - May 2027
CGPA: 8.70/10

PROJECTS

EPC Data Centre Intelligence Platform | *Next.js, FastAPI, KuzuDB, ChromaDB, Google Gemini API* | [Link](#)

- Developed a graph-centric, 5-agent full-stack platform connecting EPC specs, schedules, procurement, and logistics across 15,000+ equipment line items..
- Implemented KuzuDB graph and ChromaDB vector search powering CPM scheduling, Level 1-5 commissioning, and supplier reroutes cutting lead times from weeks to days.

ClinicalAI - Discharge Summary Agent | *Python, FastAPI, OCR, LLM APIs, PyTorch, Pydantic* | [Link](#)

- Engineered a Python/FastAPI ReAct agent running up to 10 reasoning steps across 4 clinical tools to parse OCR'd PDF records via multi-provider LLM APIs.
- Applied Pydantic schema validation, medication/result safety checks, zero-fabrication handling, and a simulated clinician-feedback loop that reached zero normalized edit distance by iteration 2 in included examples.

ESG - Carbon Emissions Tracker | *Python, Django, React, SQLite, Pandas* | [Link](#)

- Created a full-stack ESG web app with Django REST Framework, React, SQLite, and Pandas, ingesting 3 CSV data sources through 8 REST API endpoints.
- Automated Scope 1, Scope 2, and Scope 3 CO2e calculations, anomaly detection and auto-resolution, bulk approvals, audit trails, and monthly dashboard reporting.

Twitch Clipper | *Python, TwitchIO, scikit-learn, NumPy, Twitch Helix API* | [Link](#)

- Built a machine-learning Twitch clip automation tool detecting chat-activity spikes and emote bursts, then generating clips after a 7-second context-capture delay.
- Combined rule-based triggers with Isolation Forest anomaly detection and viewer-scaled thresholds, auto-saving channel-specific models every 10 minutes via Twitch Helix and WebSocket APIs.

OPEN-SOURCE CONTRIBUTIONS

- **Hugging Face Transformers** - Fixed `inverse_sqrt` scheduling so scheduler-specific kwargs, including `timescale`, are forwarded correctly.
- **Prime Intellect AI Community Environments** - Added an MMMU-Pro multimodal-reasoning benchmark environment compatible with verifiers, including data loading, reward logic, and evaluation outputs.
- **Nandatown** - Submitted a delegatable authentication plugin with HMAC-derived token chains, scope narrowing, cascading revocation, optional presenter binding, and tests.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Frameworks: FastAPI, Django, React, Next.js, Tailwind CSS

Machine Learning, AI, and Data: Transformers, Large Language Models, Optical Character Recognition, Retrieval-Augmented Generation, PyTorch, scikit-learn, Pandas, NumPy, Pydantic

Databases, APIs, and Tools: SQLite, KuzuDB, ChromaDB, Google Gemini API, WebSockets, Git, GitHub, Render

CERTIFICATIONS

- **Amazon Machine Learning Summer School** – Amazon, 2026
- **Applied Machine Learning in Python** - University of Michigan (Coursera)
- **AWS Technical Essentials** - Amazon Web Services
- **Cloud Computing** - NPTEL / IIT Kharagpur (75%), 2025

EXTRACURRICULAR

Basketball Team

Captain

Jul. 2024 - Aug. 2024
Vellore Institute of Technology

- Captained a 6-player intra-college basketball squad, demonstrating leadership and decision-making under competitive pressure
- Mentored teammates as team Captain, organizing practices and fostering communication and collaboration across the squad.